

Linearity of the Integral over Nonnegative Simple Functions

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Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and f, g be nonnegative simple functions on Ω . Then, $I(f + g) = I(f) + I(g)$ and $I(kf) = kI(f)$ for $k \geq 0$.

Proof. We first show that $I(f + g) = I(f) + I(g)$. Let $f = \sum_{j=1}^n c_j \chi_{E_j}$ and $g = \sum_{k=1}^m d_k \chi_{F_k}$, where $\{E_j\}_{j=1}^n$ and $\{F_k\}_{k=1}^m$ are partitions of Ω . Then,

$$f + g = \sum_{j=1}^n c_j \chi_{E_j} + \sum_{k=1}^m d_k \chi_{F_k}.$$

For any $1 \leq j \leq n$, we have $E_j = \bigsqcup_{k=1}^m E_j \cap F_k$ and so $\chi_{E_j} = \sum_{k=1}^m \chi_{E_j \cap F_k}$. Similarly, for any $1 \leq k \leq m$, $\chi_{F_k} = \sum_{j=1}^n \chi_{E_j \cap F_k}$. Hence,

$$\begin{aligned} f + g &= \sum_{j=1}^n c_j \sum_{k=1}^m \chi_{E_j \cap F_k} + \sum_{k=1}^m d_k \sum_{j=1}^n \chi_{E_j \cap F_k} \\ &= \sum_{j=1}^n \sum_{k=1}^m (c_j + d_k) \chi_{E_j \cap F_k} \end{aligned}$$

which is a simple function, and thus,

$$\begin{aligned} I(f + g) &= \sum_{j=1}^n \sum_{k=1}^m (c_j + d_k) \mu(E_j \cap F_k) \\ &= \sum_{j=1}^n \sum_{k=1}^m c_j \mu(E_j \cap F_k) + \sum_{k=1}^m \sum_{j=1}^n d_k \mu(E_j \cap F_k) \\ &= \sum_{j=1}^n c_j \sum_{k=1}^m \mu(E_j \cap F_k) + \sum_{k=1}^m d_k \sum_{j=1}^n \mu(E_j \cap F_k). \end{aligned}$$

We now use the additivity of μ to obtain $\sum_{k=1}^m \mu(E_j \cap F_k) = \mu(\bigsqcup_{k=1}^m E_j \cap F_k) = \mu(E_j)$ and similarly, $\sum_{j=1}^n \mu(E_j \cap F_k) = \mu(F_k)$. Hence,

$$I(f + g) = \sum_{j=1}^n c_j \mu(E_j) + \sum_{k=1}^m d_k \mu(F_k) = I(f) + I(g)$$

as desired.

Further, $kf = \sum_{j=1}^n kc_j\chi_{E_j}$, which is a nonnegative simple function, since $k \geq 0$. Thus, $I(kf) = \sum_{j=1}^n kc_j\mu(E_j) = k \sum_{j=1}^n c_j\mu(E_j) = kI(f)$.

QED